

Markscheme

Mathematics: analysis and approaches

Practice question bank

3. (a)

Consider $a = 8.1 \times 10^{-3}$ and $b = 3 \times 10^6$.

Calculate $C = \sqrt[3]{\frac{a}{b}}$, giving your answer in the form $k \times 10^n$ correct to 3 significant figures.

attempt to find $\frac{a}{b}$ (M1)

$$\frac{a}{b} = \frac{8.1 \times 10^{-3}}{3 \times 10^6} = 2.7 \times 10^{-9} \quad \text{A1}$$

attempt to take the cube root (M1)

$$C = \sqrt[3]{2.7 \times 10^{-9}} = \sqrt[3]{2.7} \times 10^{-3} \quad \text{A1}$$

$$C \approx 1.39 \times 10^{-3} \quad \text{A1}$$

Note: Do not accept a final answer given in calculator notation, e.g. $1.39E-3$.

[5 marks]

(b)

Hence write C^2 in the form $k \times 10^n$, correct to 3 significant figures.

$$C^2 = (1.39 \times 10^{-3})^2 \quad \text{(M1)}$$

$$\approx 1.94 \times 10^{-6} \quad \text{A1}$$

[2 marks]